OECD/JRC · TAI CASE

PCA와 복합지표 구축

구조 진단 · 차원축소 · 가중치 · 민감도

8 → 4

하위지표 주성분

출처: Nardo, Michela, Michaela Saisana, Andrea Saltelli, Stefano Tarantola, Anders Hoffmann, and Enrico Giovannini. 2005. Handbook on Constructing Composite Indicators: Methodology and User Guide. OECD Statistics Working Papers 2005/03. Paris: OECD Publishing. https://doi.org/10.1787/533411815016.

논문 개괄

목적

국가 비교형 복합지표의 설계 · 검증 · 소통 지침

품질 개선 · 오해 방지 · 투명성 확보

TAI CASE

23개국 × 8지표

분석 순서 · 10 STEP

1–2	3–4	5	6	7–8	9–10
이론 · 변수	구조 · 결측	정규화	가중 · 집계	강건성 · 연계	세부 · 소통

결론

단일 정답 없음

목적 적합성 · 투명성 · 분해 가능성

주요 인사이트

방법 선택 → 가중치 · 순위 변화

복합지표 = 정책결론이 아닌 토론의 출발점

이론적 틀 → 방법 선택 공개 → 강건성 검증 → 세부지표 환류

복합지표 구축 속 PCA 활용



PCA = 구조 진단 + 차원축소 + 중복정보 조정

TAI 데이터 구조

23 × 8

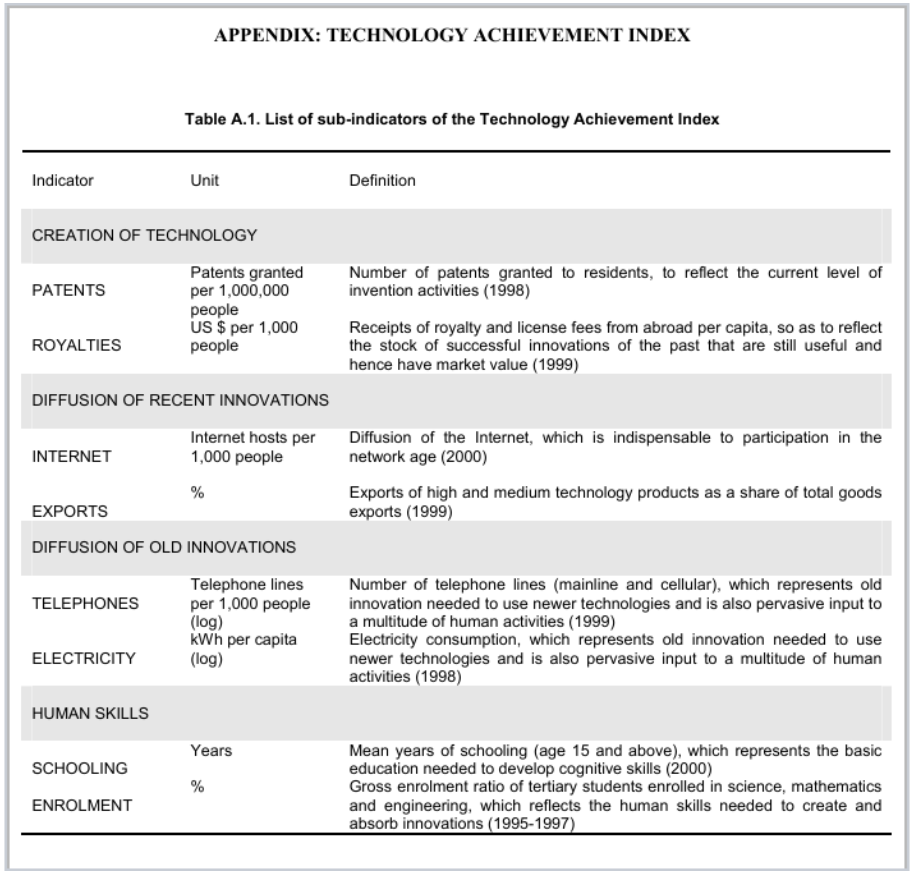
국가 × 하위지표

기술 창출
특허 · 로열티

최근 혁신 확산
인터넷 · 기술수출

과거 혁신 확산
전화 · 전력

인적 역량
학교교육 · 대학등록



APPENDIX: TECHNOLOGY ACHIEVEMENT INDEX

Table A.1. List of sub-indicators of the Technology Achievement Index

Indicator	Unit	Definition
CREATION OF TECHNOLOGY		
PATENTS	Patents granted per 1,000,000 people	Number of patents granted to residents, to reflect the current level of invention activities (1998)
ROYALTIES	US $ per 1,000 people	Receipts of royalty and license fees from abroad per capita, so as to reflect the stock of successful innovations of the past that are still useful and hence have market value (1999)
DIFFUSION OF RECENT INNOVATIONS		
INTERNET	Internet hosts per 1,000 people	Diffusion of the Internet, which is indispensable to participation in the network age (2000)
EXPORTS	%	Exports of high and medium technology products as a share of total goods exports (1999)
DIFFUSION OF OLD INNOVATIONS		
TELEPHONES	Telephone lines per 1,000 people (log)	Number of telephone lines (mainline and cellular), which represents old innovation needed to use newer technologies and is also pervasive input to a multitude of human activities (1999)
ELECTRICITY	kWh per capita (log)	Electricity consumption, which represents old innovation needed to use newer technologies and is also pervasive input to a multitude of human activities (1998)
HUMAN SKILLS		
SCHOOLING	Years	Mean years of schooling (age 15 and above), which represents the basic education needed to develop cognitive skills (2000)
ENROLMENT	%	Gross enrolment ratio of tertiary students enrolled in science, mathematics and engineering, which reflects the human skills needed to create and absorb innovations (1995-1997)

서로 다른 단위 → 표준화 → 상관행렬 PCA

원문 · p. 103 · Table A.1

중복정보 확인을 위한 PCA

r = 0.84

INTERNET ↔ ELECTRICITY

표준화

상관행렬

이중계산 가능성

Table 5. Correlation matrix for TAI sub-indicators

	PATENTS	ROYALTIES	INTERNET	EXPORTS	TELEPHONE	ELECTRICITY	SCHOOLING	ENROLMENT
PATENTS	1.00	0.13	-0.09	**0.45**	0.28	0.03	0.22	0.08
ROYALTIES		1.00	**0.46**	0.25	**0.56**	0.32	0.30	0.06
INTERNET			1.00	**-0.45**	**0.56**	**0.84**	**0.63**	0.27
EXPORTS				1.00	0.00	-0.36	-0.35	-0.03
TELEPHONE					1.00	**0.64**	0.30	0.33
ELECTRICITY						1.00	**0.65**	0.26
SCHOOLING							1.00	0.08
ENROLMENT								1.00

Note: n=23. Marked correlations are statistically significant at p < 0.05.

높은 상관 = 통계적 중복 가능성 / 정책적 동일성은 아님

원문 · p. 38 · Table 5

설명분산비율 검토

87.2%

PC1–PC4 누적

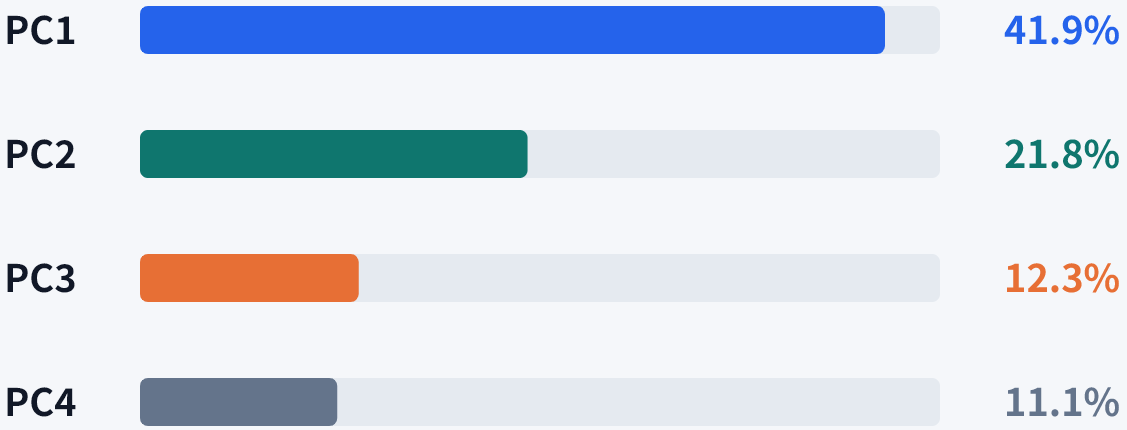


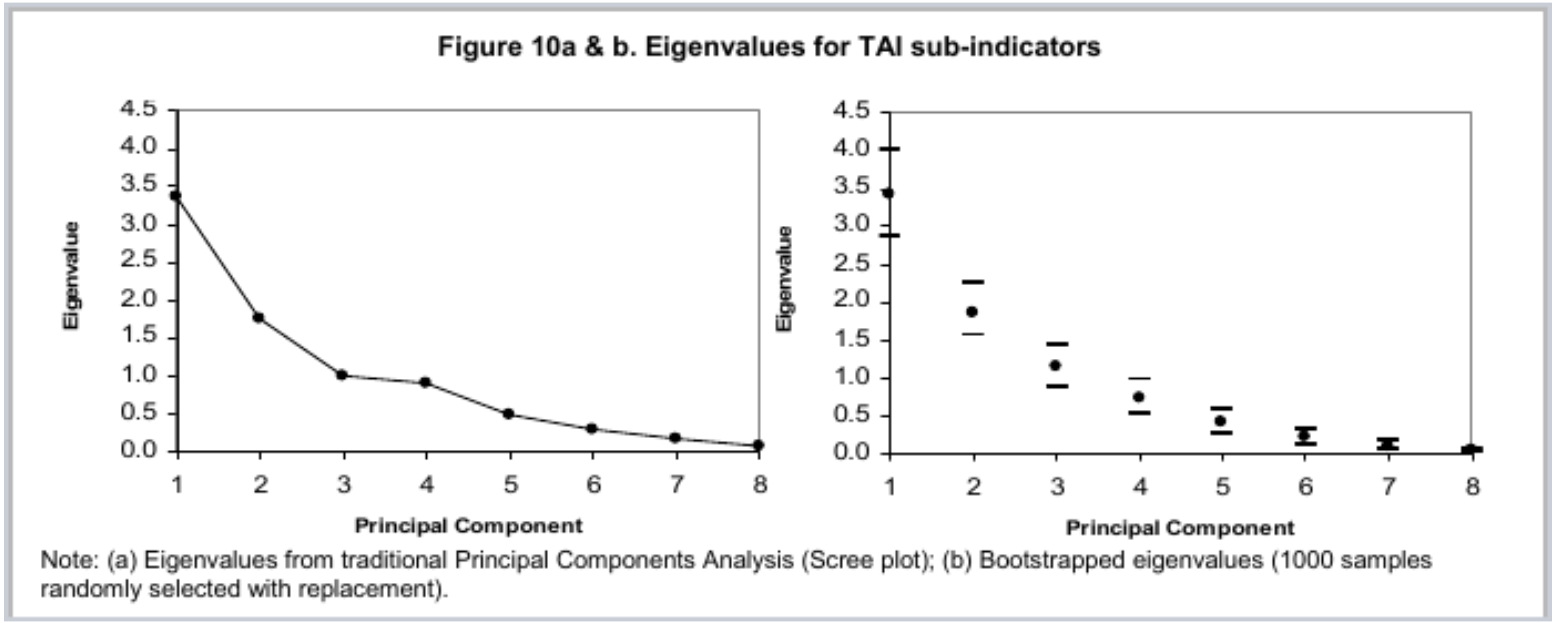
Table 6. Eigenvalues of TAI sub-indicators

	Eigenvalue	% of variance	Cumulative %
1	3.3	41.9	41.9
2	1.7	21.8	63.7
3	1.0	12.3	76.0
4	0.9	11.1	87.2
5	0.5	6.0	93.2
6	0.3	3.7	96.9
7	0.2	2.2	99.1
8	0.1	0.9	100.0

Note: Extraction method: Principal Components Analysis, n=23.

8개 지표 → 4개 성분 / 분산 보존 ≠ 개념 타당성

부트스트랩을 통한 성분 안정성 검토



n = 23

1,000회
복원추출

5–95%
고유값 구간

성분 수 = 고유값 + scree + bootstrap + 해석 가능성

적재량을 통한 성분 해석

Table 7. Component loadings for TAI sub-indicators

	1	2	3	4	5	6	7	8
PATENTS	-0.11	**-0.75**	0.13	**0.60**	-0.10	-0.12	-0.17	0.05
ROYALTIES	**-0.56**	-0.48	0.22	**-0.54**	0.27	-0.17	-0.04	0.10
INTERNET	**-0.92**	0.21	0.02	-0.10	0.04	0.11	-0.27	-0.13
EXPORTS	0.35	**-0.85**	0.01	-0.13	0.11	0.35	0.06	-0.08
TELEPHONES	**-0.76**	-0.39	-0.16	-0.16	-0.41	-0.16	0.16	-0.09
ELECTRICITY	**-0.91**	0.13	0.01	0.07	-0.19	0.30	0.04	0.16
SCHOOLING	**-0.74**	0.11	0.37	0.39	0.33	-0.02	0.20	-0.07
ENROLMENT	-0.36	-0.12	**-0.87**	0.15	0.26	-0.03	0.02	0.02

Note: Extraction method: PCA. Loadings greater than 0.5 (absolute values) are highlighted, n=23 countries.

PC1

인터넷 · 전력 · 전화 · 학교교육

PC2

특허 · 기술수출

PC3

대학 이공계 등록률

PC4

특허 · 로열티

교차적재 PATENTS · ROYALTIES

적재량 = 변수 ↔ 성분 상관 / 부호 방향은 임의

원문 · p. 42 · Table 7

회전을 통한 해석구조 단순화

PCA + VARIMAX

Table 8. Rotated factor loadings for TAI sub-indicators (method 1)

	Factor 1	Factor 2	Factor 3	Factor 4	Communality
PATENTS	0.07	**0.97**	0.06	0.06	0.95
ROYALTIES	0.13	0.07	-0.07	**0.93**	0.89
INTERNET	**0.79**	-0.21	0.21	0.42	0.89
EXPORTS	-0.64	**0.56**	-0.04	0.36	0.86
TELEPHONES	0.37	0.17	0.38	**0.68**	0.77
ELECTRICITY	**0.82**	-0.04	0.25	0.35	0.85
SCHOOLING	**0.88**	0.23	-0.09	0.09	0.85
ENROLMENT	0.08	0.04	**0.96**	0.04	0.93
Explained variance	2.64	1.39	1.19	1.76	
Cumulative (%)	33	50	65	87	

Note: Extraction method: principal components, varimax normalised rotation. Positive loadings greater than 0.5 are highlighted.

87%

ML FA + VARIMAX

Table 9. Rotated factor loadings for TAI sub-indicators (method 2)

	Factor 1	Factor 2	Factor 3	Factor 4	Communality
PATENTS	0.01	0.11	**0.88**	0.13	0.80
ROYALTIES	**0.96**	0.14	0.09	0.18	0.99
INTERNET	0.31	**0.56**	-0.29	**0.60**	0.86
EXPORTS	0.29	-0.45	**0.58**	-0.14	0.65
TELEPHONES	0.41	0.13	0.18	**0.73**	0.75
ELECTRICITY	0.13	0.57	-0.13	**0.73**	0.89
SCHOOLING	0.14	**0.95**	0.10	0.14	0.95
ENROLMENT	-0.01	0.03	0.03	0.39	0.15
Explained Variance	1.31	1.80	1.27	1.67	
Cumulative (%)	16	39	55	76	

Note: Extraction method: principal factors maximum likelihood, varimax normalised rotation.

76% · ENROLMENT 0.15

추출방법 변화 → 요인구조 · 가중치 · 순위 변화

원문 · pp. 44-45 · Tables 8-9

PCA와 복합지표 가중치 구성

적재량 → 제곱 → **중간지표** → 설명분산 결합

Table 16. Factor loadings of TAI based on principal components

	Factor 1	Factor 2	Factor 3	Factor 4	*Factor 1*	*Factor 2*	*Factor 3*	*Factor 4*
Patents	0.07	0.97	0.06	0.06	*0.00*	***0.68***	*0.00*	*0.00*
Royalties	0.13	0.07	-0.07	0.93	*0.01*	*0.00*	*0.00*	***0.49***
Internet	0.79	-0.21	0.21	0.42	***0.24***	*0.03*	*0.04*	*0.10*
Tech exports	-0.64	0.56	-0.04	0.36	*0.16*	***0.23***	*0.00*	*0.07*
Telephones	0.37	0.17	0.38	0.68	*0.05*	*0.02*	*0.12*	***0.26***
Electricity	0.82	-0.04	0.25	0.35	***0.25***	*0.00*	*0.05*	*0.07*
Schooling	0.88	0.23	-0.09	0.09	***0.29***	*0.04*	*0.01*	*0.00*
University	0.08	0.04	0.96	0.04	*0.00*	*0.00*	***0.77***	*0.00*

정책 중요도

✕

중복정보 조정

○

PCA 가중치 = 공통변동 기여 / 정책 우선순위가 아님

원문 · pp. 64-65 · Table 16

가중치 방법별 국가순위 비교

KOREA

5 → 2 FA → AHP

NORWAY

9 → 16 FA → AHP

SINGAPORE

11 → 6 FA → AHP

Table 22. TAI weights based on different methods

Equal weighting (EW), factor analysis (FA), budget allocation (BAL), and analytic hierarchy process (AHP)

Methods/ Sub-indicators	Patents	Royalties	Internet	Tech exports	Telephones	Electricity	Schooling	University
EW	0.13	0.13	0.13	0.13	0.13	0.13	0.13	0.13
FA	0.17	0.15	0.11	0.06	0.08	0.13	0.13	0.17
BAL	0.11	0.11	0.11	0.18	0.10	0.06	0.15	0.18
AHP	0.09	0.10	0.07	0.21	0.05	0.06	0.18	0.25

Table 23. TAI country rankings based on different weighting methods

Weighting method/Country	EW	FA	BOD	BAL	AHP
Finland	1	1	1	1	1
United States	2	2	1	2	3
Sweden	3	3	1	3	4
Japan	4	4	4	5	5
Korea, Rep. of	5	5	6	4	2
Netherlands	6	6	7	8	11
United Kingdom	7	8	9	7	7
Singapore	8	11	12	6	6
Canada	9	10	8	10	10
Australia	10	7	10	11	9
Germany	11	12	11	9	8
Norway	12	9	5	13	16
Ireland	13	14	13	12	12
Belgium	14	15	15	14	13
New Zealand	15	13	14	17	18
Austria	16	16	16	15	15
France	17	17	17	16	14

PCA/FA 가중치도 국가순위를 바꾸는 하나의 모형 선택

원문 · p. 73 · Tables 22-23

집계방식과 가중치 민감도 검토

집계방식 미고정

집계방식 영향 최대

선형집계 고정

가중치 총효과 0.623

Table 30. Sobol' sensitivity measures and average shift in TAI rankings

Input Factors	*First order* (S_i)	*Total effect* (S_{Ti})	$S_{Ti} - S_i$
Imputation	0.001	0.005	0.004
Normalisation	0.000	0.021	0.021
Exclusion/Inclusion of sub-indicator	0.135	0.214	0.078
Weighting Scheme	0.212	0.623	0.410
Expert selection	0.202	0.592	0.390
Sum	0.550	1.453	

Note: Significant values are underlined.

0.623

WEIGHTING · TOTAL EFFECT

PCA 결과 해석은 최종 집계함수와 민감도 분석까지 포함

원문 · pp. 81-93 · Table 30

PCA 활용 시 검증항목

표본 크기
23개국 · 8개 변수

성분 수
Kaiser · scree · 분산비율

이상치
Mahalanobis distance

안정성
bootstrap 1,000회

상관 적합성
KMO · Bartlett · VIF

해석성
적재량 · 교차적재 · 공통성

분포 가정
기술적 PCA ≠ 추론적 FA

방법 민감도
PCA · ML FA · 회전

분산 보존 ≠ 개념 타당성

표본 · 전처리 · 성분 수 · 추출방법 · 집계방식 동시 검증

원문 · pp. 40-43 · Box 3-4

PCA 활용의 해석 원칙

진단
상관 · 중복정보

→

축소
8 → 4 · 87.2%

→

구성
적재량 · 중간지표

→

검증
bootstrap · 민감도

통계적 구조 ≠ **정책적 중요도**

PCA 결과는 이론적 틀과 함께 해석